

भारतीय मानक
Indian Standard

IS 302-2-75 : 2018
(Reaffirmed 2023)

घरेलू और समान विद्युतीय उपकरणों की सुरक्षा

भाग 2 विशेष अपेक्षाएँ

खंड 75 वाणिज्यिक परिसंपक उपकरण एवं वेंडिंग मशीनें

Safety of Household and Similar Electrical Appliances

Part 2 Particular Requirements

Section 75 Commercial Dispensing Appliances and
Vending Machines

ICS 13.110;97.030

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भारतीय मानक ब्यूरो

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FOREWORD

This Indian Standard (Part 2/Sec 75) was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electro technical Division Council.

This standard covers the safety requirements of commercial dispensing appliances and vending machines.

It has been assumed in the drafting of this standard that the execution of its provisions is entrusted to appropriately qualified and experienced persons.

This standard recognizes the internationally accepted level of protection against hazards such as electrical, mechanical, thermal, fire and radiation of appliances when operated as in normal use taking into account the manufacturer's instructions. It also covers abnormal situations that can be expected in practice and takes into account the way in which electromagnetic phenomena can affect the safe operation of appliances.

This standard takes into account the requirements of IS 732 : 1989 'Code of practice for electrical wiring installations (*third revision*)' as far as possible so that there is compatibility with the wiring rules when the appliance is connected to the supply mains. However, in case of any deviation, wiring rules take precedence.

If an appliance within the scope of this standard also incorporates functions that are covered by another Part 2 of IS 302, the relevant Part 2 is applied to each function separately, as far as is reasonable. If applicable, the influence of one function on the other is taken into account.

When Part 2 standard does not include additional requirements to cover hazards dealt with in Part 1, Part 1 applies.

NOTE — This means that in such a case, it has been decided that for the Part 2 standards that it is not necessary to specify particular requirements for the appliance in question over and above the general requirements.

This standard is a product family standard dealing with the safety of appliances and takes precedence over horizontal and generic standards covering the same subject.

NOTE — Horizontal and generic standards covering a hazard are not applicable since they have been taken into consideration when developing the general and particular requirements for the IS 302 series of standards.

An appliance that complies with the text of this standard will not necessarily be considered to comply with the safety principles of the standard if, when examined and tested, it is found to have other features which impair the level of safety covered by these requirements.

An appliance employing materials or having forms of construction differing from those detailed in the requirements of this standard may be examined and tested according to the intent of the requirements and, if found to be substantially equivalent, may be considered to comply with the standard.

This standard is to be read in conjunction with the latest edition of IS 302-1 'Safety of household and similar electrical appliances : Part 1 General requirements' and its amendments. This standard was prepared on the basis of IS 302-1: 2008.

NOTE — When 'Part 1' is mentioned in this standard, it refers to IS 302-1.

This Part 2 supplements or modifies the corresponding clauses in IS 302-1, so as to convert that standard into the Indian Standard: Particular requirements for Commercial Dispensing appliances and Vending machines.

When a particular subclause of Part 1 is not mentioned in this Part 2, that subclause applies as far as is reasonable. When this standard states "addition", "modification" or "replacement", the relevant text in Part 1 is to be adapted accordingly.

NOTE — The following numbering system is used:

- a) subclauses, tables and figures that are numbered starting from 101 are additional to those in Part 1;
- b) unless notes are in a new subclause or involve notes in Part 1, they are numbered starting from 101, including those in a replaced clause or subclause; and
- c) additional annexes are lettered AA, BB, etc.

This standard is based on IEC 60335-2-75 : 2012 (Ed. 3.0). As this standard refers to IS 302-1, the differences of IS 302-1 from IEC 60335-1 would apply. Leakage current limits in this Part have been modified to align with those in IS 302-1.

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 75 Commercial Dispensing Appliances and Vending Machines

1 SCOPE

This clause of Part 1 is replaced by the following:

This Indian Standard deals with the safety of electric commercial dispensing appliances and vending machines for preparation or delivery of food, drinks and consumer products, their rated voltage being not more than 250 V for single-phase appliances and 480 V for other appliances.

NOTE 101 — Examples of appliances that are within the scope of this standard are:

- a) bulk tea or coffee brewing machines;
- b) cigarette vending machines;
- c) coffee grinders;
- d) commercial liquid heaters;
- e) coffee makers with or without integrated coffee grinder;
- f) coffee makers with cooling systems;
- g) hot and cold beverage vending machines;
- h) hot water dispensers;
- j) ice cream and whipped cream dispensers;
- k) ice dispensers;
- m) newspaper, audio or video tape or disc vending machines;
- n) packaged food and drink vending machines; and
- p) refrigerated merchandisers.

Appliances can have more than one function.

NOTE 102 — Other standards may be applicable for some functions such as

- a) refrigeration (IS 302-2-24);
- b) heating by microwaves (IS 302-2-25).

This standard also deals with the hygiene aspects of appliances.

As far as is practicable, this standard deals with the common hazards presented by appliances that are encountered by users and maintenance persons. However, in general, it does not take into account young children playing with the appliance.

NOTE 103 — Attention is drawn to the fact that:

- a) for appliances intended to be used in vehicles or on board ships or aircraft, additional requirements may be necessary;
- b) in many countries, additional requirements for appliances incorporating pressure vessels are specified; and
- c) in many countries, additional requirements are specified by

the national health authorities, the national authorities responsible for the protection of labour, the national water supply authorities and similar authorities.

NOTE 104 — This standard does not apply to

- a) appliances intended to be used exclusively for household purposes;
- b) appliances intended to be used exclusively for industrial purposes;
- c) appliances intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas);
- d) commercial electric boiling pans;
- e) commercial electric bains-marie;
- f) amusement machines and personal service machines;
- g) appliances solely used for dispensing money;
- h) display cabinets; and
- j) appliances incorporating electrode-type water heaters.

2 REFERENCES

This clause of Part 1 is applicable except as follows.

Addition:

<i>IS No.</i>	<i>Title</i>
1401 : 2008	Protection of persons and equipment by enclosures — Probes for verification
3400 (Part 6) : 2018	Methods of test for vulcanized rubbers: Part 6 Determination of the effect of liquids
12032 (Part 2) : 1987	Graphical symbols for diagrams in the field of electrotechnology: Part 2 Symbols elements, qualifying symbols and other symbols having general application
IS/IEC 60320-1 : 2001	Appliance couplers for household and similar general purposes : Part 1 General requirements
IS/IEC 61058-1: 2000	Switches for appliances : Part 1 General requirements

3 TERMS AND DEFINITIONS

This clause of Part 1 is applicable except as follows:

3.1.9 Replacement:

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Normal Operation — Operation of the appliance under the following conditions:

The appliance is operated in the standby mode until steady conditions are established and then under the most unfavourable dispensing procedure. The appliance is refilled when necessary in accordance with the instructions for use, or the instructions for maintenance, and the next operating period started as soon as possible.

Lids and covers of appliances of the professional type and of appliances of the supervised type are placed in their intended positions.

Coffee makers are operated with their container filled with the rated capacity of water, or connected to the water mains, if applicable. Coffee makers with a heated surface intended to keep the liquid warm are operated with or without the container, whichever is the more unfavourable.

3.6.2 Replacement:

Detachable Part — Part that can be removed without the aid of a tool, a part that is removed in accordance with the instructions for use or the instructions for maintenance, even if a tool or access key is needed for removal, or a part that does not fulfill the test of 22.11.

NOTES

- 1 If a part has to be removed for installation purposes, this part is not considered to be detachable even if the instructions state that it is to be removed.
- 2 A part that can be opened is considered to be a part that can be removed.

3.7.3 Replacement:

Thermal cut-out — Device that during abnormal operation limits the temperature of the controlled part by automatically opening the circuit, or reducing the current, and is constructed so that its setting cannot be altered by the user or the maintenance person.

3.8.5 Replacement:

Maintenance Operation — Operation performed in the maintenance area or user area, such as preparing the appliance for new products or new operating methods, cleaning, price changing, replenishing, coin collecting and setting of controls or similar operations.

NOTE — Maintenance operation does not include operations performed in the service area.

3.101 Rated Pressure — Pressure assigned to the pressurized parts of the appliance by the manufacturer.

3.102 Standby Mode — State of the appliance when filled as intended with ingredients or products, energized and ready for use, cash boxes and overflow containers being empty.

3.103 Access Key — Key or other means that gives access to the maintenance area but does not give access to the service area.

NOTE — “Other means” includes a *tool* or operation by codes or signals produced by optical or electro-magnetic sources.

3.104 Override Key — Key or other means that is used to render an interlock inoperative.

3.105 Dispensing Appliance — Appliance intended to deliver or make available food, drinks or other consumer products.

NOTES

- 1 The appliance may also prepare the products.
- 2 The dispensing operation may be initiated manually or by means such as coins or credit cards.

3.106 Vending Machine — Dispensing appliance that is operated by coins, credit cards or other means of payment.

3.107 Instructions For Maintenance — Instructions explaining how to carry out cleaning, replenishing, coin collecting, setting of controls and similar operations.

3.108 Maintenance Person — Person who maintains the appliance in accordance with the instructions for maintenance.

3.109 User Area — Area where access is gained without the use of an access key or a tool.

NOTES

- 1 The user area of appliances of the supervised type is determined with detachable parts and other movable parts, such as doors and lids, in position as in normal use.
- 2 Appliances of the professional type have no user area.

3.110 Maintenance Area — Area where access can only be gained by the use of an access key.

3.111 Service Area — Area where access cannot be gained by the use of an access key alone.

3.112 Appliance of the Professional Type — Dispensing appliance that is only intended to be used by trained personnel such as kitchen or bar staff.

3.113 Appliance of the Supervised Type — Dispensing appliance that is intended to be maintained by trained personnel but may be used by other persons in a location where its use is overseen.

NOTE — Dining rooms in restaurants are examples of such locations.

3.114 Potentially Hazardous Food — Food which includes natural or synthetic ingredients that are capable of supporting rapid and progressive growth of pathogenic or toxin producing micro-organisms.

NOTES

- 1 Examples of potentially hazardous food are milk, eggs, meat, poultry, shellfish, crustaceans, and their products, either raw or heat treated, as well as food of plant origin that is ready for

consumption without the need for any further preparation or processing.

2 Food may become potentially hazardous food during processing, for example when powdered ingredients are mixed with water or when food is stored at incorrect temperature.

3 Potentially hazardous food does not include:

- a) candy, nuts, gum and similar confectionery;
- b) cookies, crackers and similar bakery products;
- c) instant-coffee, chocolate, cocoa and sugar;
- d) food having a pH level of not greater than 4.6 or a water activity (Aw) value not greater than 0.85 at 25 °C;
- e) food maintained at a temperature not exceeding 5 °C for periods specified by the producer, but for not more than 5 days;
- f) food maintained at a temperatures above 65 °C or below -18 °C;
- g) food in hermetically sealed containers; and
- h) food that has been processed to prevent spoilage.

3.115 Espresso Coffee Maker — Coffee maker in which water is heated and forced through the ground coffee by steam pressure or by means of a pump.

NOTE — Espresso coffee makers can have an outlet for supplying steam or hot water.

4 GENERAL REQUIREMENT

This clause of Part 1 is applicable.

5 GENERAL CONDITIONS FOR THE TESTS

This clause of Part 1 is applicable except as follows.

5.2 Addition:

If the test of **15.102** has to be carried out, three additional samples are required.

5.6 Replacement:

Controls or switching devices in the user area are adjusted to the most unfavourable setting.

Controls, switching devices or other parts in the maintenance area are adjusted to the most unfavourable setting within limits stated in the instructions for maintenance. Controls or switching devices in the service area are not adjusted.

5.9 Addition:

When alternative software is made available by the appliance manufacturer, the appliance is tested with the software that gives the most unfavourable results.

5.10 Addition:

NOTE 101 — Access keys and override keys can be supplied separately from the appliance.

Appliances are installed in accordance with the instructions provided with the appliance before testing.

If the instructions state that the appliance may be installed together with other appliances, the effect of this combination is taken into account.

5.101 Appliances intended to be connected to the water mains are supplied with water having a temperature of $15\text{ °C} \pm 5\text{ °C}$ and the most unfavourable pressure specified in the instructions. For appliances that are manually filled with water, the temperature of the water is $15\text{ °C} \pm 5\text{ °C}$.

For appliances intended to cool water, the temperature of the water is $25\text{ °C} \pm 5\text{ °C}$.

5.102 The requirements of this standard for the maintenance area are applicable when the instructions for maintenance are being followed. If an access key is provided for access to the maintenance area, it is used before a test is carried out if this is more unfavourable.

5.103 When reference is made to the application of test probe B, test probe 18 of IS 1401 is also applied in the user area.

5.104 Appliances of the professional type and appliances of the supervised type are tested as heating appliances even if they incorporate a motor. If these appliances do not contain heating elements, they are tested as motor-operated appliances.

6 CLASSIFICATION

This clause of Part 1 is applicable except as follows.

6.1 Modification:

Appliances shall be class I, class II or class III.

6.2 Addition:

Appliances intended for outdoor use shall be at least IPX4.

Appliances that may be cleaned by water jets, or installed where water jets are liable to be used, shall be at least IPX5.

7 MARKING AND INSTRUCTIONS

This clause of Part 1 is applicable except as follows.

7.1 Addition:

Appliance shall be marked with:

- a) their rated pressure, in megapascals, if applicable; and
- b) the maximum permissible water pressure, in megapascals, for appliances intended to be connected to the water mains.

Appliances intended to be filled by hand shall have means that indicate when the required level for correct operation has been reached.

NOTE 101 — A level mark or an audible or visual signal are suitable means.

For appliances incorporating a socket outlet, the voltage, nature of the supply and current or power

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output shall be marked in the vicinity of the socket outlet.

Appliances intended to be partially immersed in water for cleaning shall be marked with the maximum level of immersion and with the substance of the following:

Do not immerse beyond this level.

7.3 Addition:

The requirement also applies when the adjustment has to be made by the maintenance person.

7.8 Addition:

Terminals for equipotential bonding shall be indicated by symbol 02-15-05 of IS 12032 (Part 2).

This symbol shall not be placed on screws, removable washers or other parts that can be removed when conductors are being connected.

7.12.1 Addition:

The installation instructions for appliances intended to be connected to the water mains shall specify the means of connection and draw attention to any national rules that may be applicable.

The installation instructions shall state if the appliance is suitable for outdoor use.

The installation instructions shall state the maximum and minimum ambient temperatures for correct operation.

For appliances that are not at least IPX5, the instructions shall state that the appliance is not suitable for installation in an area where a water jet could be used.

The installation instructions shall state the maximum tilt of the appliance for safe operation. A tilt of less than 2° need not be stated. An instruction such as “the appliance has to be placed in a horizontal position” is sufficient.

The installation instructions for appliances of the professional type shall state that the appliance is only to be installed in locations where its use and maintenance is restricted to trained personnel.

The installation instructions for appliances of the supervised type shall state that the appliance is only to be installed in locations where it can be overseen by trained personnel.

The installation instructions for class I appliances of the professional type that are intended to be permanently connected to fixed wiring, and have a leakage current that may exceed 10 mA, shall state that the installation of a residual current device (RCD) having a rated residual operating current not exceeding 30 mA is advisable.

7.12.101 If it is necessary to take special precautions during maintenance operations, details of these shall be supplied. The instructions for maintenance shall state how to gain access to the maintenance area including how to use the access key and the override key. They shall not include instructions on how to gain access to a service area.

Compliance is checked by inspection.

7.12.101.1 The instructions for maintenance shall specify the method and frequency of cleaning. They shall include details for descaling, disinfecting, flushing and removal of any residual cleaners, sterilizers or descalers from the appliance, if applicable. Recommended cleaning or disinfecting agents shall be specified, and they may be identified by their chemical denomination.

If the appliance is not at least IPX5, the instructions for maintenance shall state that the appliance must not be cleaned by a water jet.

The instructions for maintenance for appliances incorporating an appliance inlet, and intended to be partially or completely immersed in water for cleaning, shall state that the connector must be removed before the appliance is cleaned and that the appliance inlet must be dried before the appliance is used again.

Compliance is checked by inspection.

7.12.101.2 If the use of an override key allows access to moving parts, a suitable warning shall be given in the instructions for maintenance.

Compliance is checked by inspection.

7.12.101.3 The instructions for maintenance shall list any accessories that may be used with the appliance.

Compliance is checked by inspection.

7.12.101.4 The instructions for maintenance shall state the maximum and minimum ambient temperatures for correct operation.

For appliances using water, the instructions for maintenance shall give details concerning the prevention of freezing or how to ensure safe operation if freezing occurs.

Compliance is checked by inspection.

7.12.101.5 The instructions for maintenance for appliances containing pressurized gas shall give details on the safe handling of the pressurized containers and of the gas.

Compliance is checked by inspection.

7.12.101.6 The instructions for maintenance shall specify the types of food for which the appliance is

suitable, and give details on how to ensure hygienic operation.

Compliance is checked by inspection.

7.12.101.7 The instructions for maintenance for appliances intended for dispensing potentially hazardous food, when the safety of the food depends upon the temperature and duration of storage, shall include details for the safe handling of the food.

Compliance is checked by inspection.

7.12.102 The instructions shall state that access to the service area is restricted to persons having knowledge and practical experience of the appliance, in particular as far as safety and hygiene are concerned.

Compliance is checked by inspection.

8 PROTECTION AGAINST ACCESS TO LIVE PARTS

This clause of Part 1 is applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of Part 1 is not applicable.

10 POWER INPUT AND CURRENT

This clause of Part 1 is applicable.

11 HEATING

This clause of Part 1 is applicable except as follows.

11.2 Modification:

Appliances normally fixed to the floor, and those having a mass greater than 40 kg and not provided with casters or rollers, are installed in accordance with the instructions. If no instructions are provided, the appliance is placed on the floor as close to the walls as possible.

Other appliances, except fixed appliances, are placed on the floor as near to the walls as possible.

11.4 Addition:

Heating appliances with electronic circuits controlling the power input are operated as combined appliances.

If the temperature rise limits are exceeded in appliances incorporating motors, transformers or electronic circuits, and if the power input is lower than the rated power input, the test is repeated with the appliance supplied at 1.06 times the rated voltage.

11.6 Addition:

Combined appliances without electronic power controls are operated as heating appliances.

11.7 Replacement:

The appliance is operated under normal operation until steady conditions are established, the appliance being refilled when necessary.

NOTE 101 — Refilling can require the use of an access key.

11.8 Addition:

The temperature rise of the surfaces in the user area shall not exceed the limits specified for handles, knobs, grips and similar parts that are held for short periods only. This does not apply to the surface of parts that need to be hot in order for the appliance to fulfil its function.

The temperature rise limits of motors, transformers and components of electronic circuits, including parts directly influenced by them, may be exceeded when the appliance is operated at 1.15 times the rated power input.

11.101 Appliances incorporating refrigerating equipment, and having motor-compressors, are also tested at an ambient temperature of

- a) 32 °C, for appliances for temperate countries; and
- b) 43 °C, for appliances for tropical countries.

Other parts of the appliance are operated to produce the most unfavourable conditions in the refrigerating system.

Temperature rises of parts of the appliance, other than the motor-compressor, are not determined.

The temperature of windings and the enclosure of motor-compressors shall not exceed the following values:

- a) 140 °C, for windings of motor-compressors with synthetic insulation;
- b) 130 °C, for windings of motor-compressors with cellulosic insulation; and
- c) 150 °C, for external enclosures of motor-compressors.

12 VOID

13 LEAKAGE CURRENT AND ELECTRIC STRENGTH AT OPERATING TEMPERATURE

This clause of Part 1 is applicable except as follows.

13.2 Modification:

For stationary class I heating appliances, the leakage current shall not exceed the following values:

- for appliances of the 1 mA per kW rated professional type intended to power input of the be permanently connected to appliance, with no fixed wiring; maximum;

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- for other appliances of the professional type 1 mA per kW rated power input of the appliance, with a maximum of 10 mA;
- for other heating appliances 0.21 mA or 0.21 mA per kW rated power input of the appliance, whichever is higher with a maximum of 1.5 mA.

14 TRANSIENT OVERVOLTAGES

This clause of Part 1 is applicable.

15 MOISTURE RESISTANCE

This clause of Part 1 is applicable except as follows.

15.1.1 Addition:

Appliances of the professional type classified IPX3 or lower, and intended to be placed on a kitchen floor, are subjected to a test in which water under pressure is indirectly splashed onto the appliance. The splash apparatus is shown in Figure 101. The bowl is placed on the floor and the water pressure adjusted so that the water splashes to a height of 150 mm above the bottom of the bowl. The apparatus is moved around the appliance in order to splash it from all directions for a total of 5 min.

15.2 Replacement:

Appliances subject to spillage of liquids or solids in normal use shall be constructed so that spillage does not affect their electrical insulation. The electrical insulation shall not be affected by cleaning, disinfecting, descaling and similar operations.

Compliance is checked by the tests of **15.2.101** to **15.2.113**.

Water used for the tests shall contain approximately 1 percent NaCl.

Appliances with type X attachment, except those having a specially prepared cord, are fitted with the lightest permissible type of flexible cord of the smallest cross-sectional area specified in Table 11 of Part 1.

Appliances incorporating an appliance inlet are tested with or without an appropriate connector in position, whichever is more unfavourable.

Before each test, the appliance is operated in the standby mode. Containers that are connected to the water mains are prefilled with saline solution.

After each overfilling or application of liquid, the

appliance shall withstand the electric strength test of **16.3** and inspection shall show that there is no trace of liquid or solids on insulation that could result in a reduction of clearances and creepage distances below the values specified in **29**. All residues are then removed and the appliance is dried.

Detachable parts in the user area are removed or placed in the most unfavourable position.

Detachable parts in the maintenance area are placed in their normal position following a maintenance operation.

15.2.101 Containers for ingredients or products in powdered or granulated form are filled with dry granulated sugar, ignoring any level indication. A further quantity equal to 15 percent of the total capacity of the container is then poured in steadily over a period of 1 min.

Containers that are intended to be filled outside the appliance are replaced without removing any excess sugar from the outside of the container. Lids are replaced after overfilling.

15.2.102 Liquid containers that are filled manually are filled with saline solution and a further quantity equal to 15 percent of the total capacity of each container or 0.25 litre, whichever is the greater, is poured in steadily over a period of 1 min.

15.2.103 The outlets of liquid mixing containers are blocked and the containers are filled with saline solution. A further quantity equal to 15 percent of the total capacity of each container or 0.25 litre, whichever is the greater, is poured in steadily over a period of 15 s. If the container has more than one independent outlet, they are blocked in turn.

15.2.104 Drains for liquid waste containers are blocked and the containers are filled with saline solution. A further quantity equal to 15 percent of the total capacity of each container or 0.25 litre, whichever is the greater, is poured in steadily over a period of 15 s. If the container has more than one independent drain, they are blocked in turn. If there is more than one container, they are tested in turn.

15.2.105 Drain taps of containers used during maintenance operations are adjusted in turn to the most unfavourable position. The appliance is supplied at rated voltage and operated under normal operation until the flow of saline solution stabilizes.

15.2.106 Failure of the inlet valve of appliances connected to the water mains is simulated. Water is allowed to flow for 1 min after the first evidence of overflow unless the inflow stops automatically. The failure of only one device is tested at a time.

15.2.107 Appliances dispensing liquid into a serving container, such as a cup or jug, are tested by rapidly pouring 0.5 litre of saline solution over the surface where the container is filled, transported and removed by the user.

15.2.108 Appliances with accessible openings, including slots for coins or cards, other than appliances of the professional type and appliances of the supervised type, are tested by slowly pouring 0.25 litre of saline solution into each opening. If the opening is in a vertical surface, the solution is projected towards the opening.

15.2.109 Appliances having external surfaces on which it is possible to place a vessel, such as a cup or jug, are tested by rapidly pouring 0.5 litre of saline solution over the surface. The quantity of saline solution is increased to 5 litre for appliances of the professional type if their highest surface is lower than 1.5 m. The test is carried out even if the appliance does not dispense liquid. If there is more than one surface, they are tested in turn.

For espresso coffee maker of the professional type, the amount of water is increased to 5 litre only if the highest surface after installation is lower than 1.2 m.

15.2.110 Appliances delivering pre-packed products are tested to simulate leakage from the package over any area where the package is stored or transported.

Leakage from liquid products is simulated by rapidly pouring a quantity of saline solution, equal in volume to the largest pre-packed product that can be delivered from the appliance, over the area.

Leakage from dry products is simulated by rapidly pouring a quantity of dry granulated sugar, equal in volume to the largest pre-packed product that can be delivered from the appliance, over the area. This test is not applicable to appliances intended to deliver only solid products such as newspapers, films or cigarettes.

15.2.111 *Maintenance operations* involving the use of liquids are carried out three times.

15.2.112 Parts liable to be cleaned are wiped with a sponge, having dimensions approximately 150 mm × 75 mm × 50 mm, saturated with saline solution. The sponge is applied without appreciable force for approximately 10 s to each surface. This test is not applied to surfaces in the maintenance area for which cleaning instructions are given.

15.2.113 Appliances subject to descaling are descaled 10 times in accordance with the instructions for maintenance. The appliance is then operated in the standby mode.

15.3 Addition:

If it is not possible to place the appliance in the humidity cabinet, electrical parts are tested separately.

15.101 Appliances having a tap that provides water for filling or cleaning shall be constructed so that the water cannot come into contact with live parts or affect electrical insulation.

Compliance is checked by the following test.

The appliance is connected to the water mains, the pressure being adjusted to the maximum water pressure marked on the appliance. Tiltable and movable parts, including lids, are placed in the most unfavourable position. The tap is fully opened for 1 min, swivel outlets being adjusted to direct the water in the most unfavourable direction. The appliance shall then withstand the electric strength test of **16.3**.

15.102 Appliances intended to be partially or completely immersed in water for cleaning shall have adequate protection against the effects of immersion.

Compliance is checked by the following tests, which are carried out on three additional appliances.

The appliances are operated under normal operation at 1.15 times rated power input, until the thermostat operates for the first time. Appliances without a thermostat are operated until steady conditions are established. The appliances are disconnected from the supply, any appliance connector being withdrawn. They are then completely immersed in water containing approximately 1 percent NaCl and having a temperature between 10 °C and 25 °C, unless they are marked with the maximum level of immersion, in which case they are immersed 50 mm deeper than this level.

After 1 h, the appliances are removed from the saline solution, dried and subjected to the leakage current test of **16.2**.

This test is carried out four more times, after which the appliances shall withstand the electric strength test of **16.3**, the voltage being as specified in Table 4 of Part 1.

The appliance having the highest leakage current after the fifth immersion is dismantled and inspection shall show that there is no trace of liquid on insulation that could result in a reduction of clearances and creepage distances below the values specified in **29**.

The remaining two appliances are operated under normal operation at 1.15 times rated power input for 240 h. After this period, the appliances are disconnected from the supply and immersed again for 1 h. They are then dried and subjected to the electric strength test

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of **16.3**, the voltage being as specified in Table 4 of Part 1.

Inspection shall show that there is no trace of liquid on insulation that could result in a reduction of clearances and creepage distances below the values specified in **29**.

16 LEAKAGE CURRENT AND ELECTRIC STRENGTH

This clause of Part 1 is applicable except as follows.

16.2 Modification:

For stationary class I heating appliances, the leakage current shall not exceed the following values:

- for appliances of 2 mA per kW rated the professional power input of the type intended to be appliance, with no permanently maximum; connected to fixed wiring;
- for other appliances 2 mA per kW rated of the professional power input of the type appliance, with a maximum of 10 mA;
- for other heating 0.21 mA or 0.21 mA appliances per kW rated power input of the appliance, whichever is higher with a maximum of 1.5 mA.

17 OVERLOAD PROTECTION OF TRANSFORMERS AND ASSOCIATED CIRCUITS

This clause of Part 1 is applicable.

18 ENDURANCE

This clause of Part 1 is not applicable.

19 ABNORMAL OPERATION

This clause of Part 1 is applicable except as follows:

19.1 Addition:

Appliances are also subjected to the tests of **19.101** and **19.102**, if applicable.

Detachable parts in the user area are removed or placed in the most unfavourable position.

Detachable parts in the maintenance area are placed in their normal position following a maintenance operation.

Containers are filled to the most unfavourable level.

Appliances having a control that limits the pressure during the tests of 11 are subjected to the tests of 19.4 with this control rendered inoperative.

19.2 Addition:

NOTE 101 — Examples of achieving restricted heat dissipation are

- a) operating without water;
- b) switching off the fan; and
- c) covering ventilation openings.

19.4 Addition:

If a control also performs other functions, only the part controlling the temperature or pressure is rendered inoperative.

19.11.2 Modification:

The fault conditions are simulated until steady conditions are established.

19.13 Addition:

During the tests, molten plastic shall not be emitted.

Liquid having a temperature above 80 °C, steam or solid objects shall not be emitted from unexpected places in a way likely to cause injury to persons.

After the tests, compliance with **15.1** and **15.2** shall not be impaired. The electric strength test of **16.3** shall be carried out after each test if it is expected that the electrical insulation could be affected.

19.101 The appliance is supplied at rated voltage and operated under normal operation. Any fault condition or unexpected operation likely to occur during use of the appliance is introduced.

If operation without water in the appliance is considered to be a more unfavourable condition, the tests are carried out with the water supply valve closed. The water supply valve is not closed during the dispensing operation.

NOTES

1 Damaged components or parts can be replaced after each test.

2 Examples of fault conditions or unexpected operation are

- a) defects in the appliance:
 - i) a programmer stopping in any position;
 - ii) disconnection and reconnection of one or more phases of the supply mains during any part of the programme;
 - iii) open-circuiting or short-circuiting of components;
 - iv) locking the main contacts of a contactor in the “on” position if they are used for energizing heating elements. However, this defect is not introduced if at least two independent sets of contacts are provided. This can be achieved by two contactors operating independently of each other or by one contactor having two independent armatures operating two independent sets of main contacts;
 - v) failure of a magnetic valve;

- vi) failure of a pneumatic or hydraulic control; and
- vii) blocking the coin or product channels. If blockage can be noticed from the outside of the appliance, further delivery is not attempted, otherwise the appliance is operated until no further delivery is possible. The wrapping of products in conductive materials has to be taken into account.
- b) faulty operation by users or maintenance persons:
 - i) incorrect actuation of knobs, handles, switches or push-buttons;
 - ii) interrupting the dispensing operation by available facilities;
 - iii) incorrect opening or closing of doors or lids;
 - iv) improper application of the instructions for maintenance;
 - v) incorrect routine cleaning. The sponge test of **15.2.112** is applied to all surfaces in the user area. It is also applied to all surfaces in the maintenance area, except those for which cleaning instructions are given;
 - vi) setting controls, switches or programmers in the most unfavourable position;
 - vii) incorrect loading; and
 - viii) incorrect coin collection.
- c) abuse by users:
 - i) obstructing dispensing openings; and
 - ii) blocking moving parts.

3 In general, tests are limited to the fault conditions that may be expected to give the most unfavourable results.

19.102 Appliances incorporating a thermal cut-out of the capillary type are tested as specified in **19.4** but with the capillary tube ruptured.

20 STABILITY AND MECHANICAL HAZARDS

This clause of Part 1 is applicable except as follows:

20.1 Modification:

The appliance is tested with doors, lids and similar parts in the maintenance area placed in the normal position of use.

The test with the appliance tilted to 15° is not carried out.

Addition:

The test is repeated with doors, lids and similar parts in the maintenance area placed in the most unfavourable position, however, the appliance is only tilted to an angle of 5°.

20.2 Addition:

Covers over moving parts having a kinetic energy exceeding 4 J shall be interlocked so that it is only possible to remove them when the parts are stationary unless they are only removable with the aid of a *tool*.

21 MECHANICAL STRENGTH

This clause of Part 1 is applicable except as follows:

21.1 Addition:

The impact energy of 0.5 J is applied in the maintenance area. In the user area, the value of the impact energy is 1.0 J.

22 CONSTRUCTION

This clause of Part 1 is applicable except as follows:

22.6 Addition:

Parts that withstand the aging test of Annex AA are not considered to be parts where leakage could occur.

Drain holes in coffee makers shall be at least 5 mm in diameter or 20 mm² in area with a width of at least 3 mm.

Compliance is checked by measurement.

22.7 Addition:

Pressure relief devices shall be constructed so that they cannot be rendered inoperative or set to a higher pressure without the aid of a tool that is normally only available to the manufacturer.

Appliances incorporating pressurized systems are subjected to the following test.

All pressure regulating devices are rendered inoperative and the system is filled with water. The pressure is then raised hydraulically until the pressure relief device operates.

The pressure shall not exceed 1.2 times the rated pressure and the appliance shall be fit for further use. The pressure relief device is then rendered inoperative and the pressure again raised until twice the rated pressure is attained. The pressure is maintained at this value for 5 min.

The system shall not rupture and there shall be no permanent deformation. However, an intentionally weak part may rupture after the pressure has attained 1.5 times the rated pressure as long as it does not give rise to a hazard. In this case, the weak part is replaced and the test repeated. Rupture shall occur in the same way.

If fluid cannot circulate freely throughout the pressurized system, separate tests can be carried out on individual parts of the system.

If more than one pressure relief device operates on the same part of the system, they are rendered inoperative together.

This test is not made on refrigerating systems.

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The appliance shall then withstand the electric strength test of **16.3**.

22.14 Addition:

The requirement also applies in the maintenance area to parts liable to be touched during maintenance operations.

22.33 Addition:

Ingredients and products shall not be in direct contact with live parts or, for class II construction, with basic insulation.

22.47 Addition:

All pressure regulating devices are rendered inoperative.

22.101 Appliances shall be constructed so that interlocks cannot be rendered inoperative without using an override key if they are necessary for compliance with the standard.

Compliance is checked by inspection, by manual test and by applying test probe B of IS 1401.

22.102 It shall not be possible to gain access to the service area by only using the access key for the maintenance area.

Compliance is checked by inspection and by manual test.

22.103 Appliances shall be constructed so that scalding by steam is prevented when a lid is opened.

Compliance is checked by inspection and by the tests of **19**.

22.104 Appliances shall be constructed so that dispensed products cannot be contaminated by substances such as lubricants and debris.

Compliance is checked by inspection.

22.105 Appliances shall be constructed so that it is not possible to inadvertently open draw-off taps and drain valves or withdraw drain plugs.

Compliance is checked by inspection and by manual test.

Valves that return automatically to the closed position when released, those of the wheel type or those placed in a recess are considered to comply with this requirement.

22.106 Coin boxes and containers for other payment means shall be positioned or protected so that overfilling cannot cause a hazard.

Compliance is checked by inspection.

22.107 Appliances intended to be connected to the water mains shall be constructed for a water pressure not less than 0.6 MPa.

Compliance is checked by inspection.

22.108 Appliances shall be protected in such a manner that moisture, grease and products used in the appliance will not accumulate so that clearance and creepage distances are affected.

Compliance is checked by inspection.

22.109 Lights indicating a warning against a hazard shall only be coloured red.

Compliance is checked by inspection.

22.110 Appliances having pressurized containers shall be constructed so that the lid cannot be removed while the pressure within the container is excessive. They shall incorporate a means to release the pressure to a value such that the lid can be removed without risk.

Compliance is checked by the following test.

The appliance is operated as specified in **11** until the pressure regulator operates for the first time.

The appliance is then disconnected from the supply and the pressure allowed to decrease until the pressure is 4 kPa. A force of 100 N is applied to the most unfavourable point where the lid or its handle can be gripped. It shall not be possible to remove the lid.

The internal pressure is then gradually reduced, the force of 100 N being maintained. There shall be no hazardous displacement of the lid when it is released.

This test is not carried out on appliances when the lid is secured by screw clamps or other devices that ensure that the pressure is automatically reduced in a controlled manner before the lid can be removed.

22.111 Appliances for dispensing potentially hazardous food shall incorporate means to prevent the dispensing of the food if it has been adversely affected by storage or process temperatures. This requirement does not apply to coffee makers of the professional type and coffee makers of the supervised type.

Compliance is checked by inspection.

22.112 Surfaces of food areas and splash areas shall be cleanable so that all unwanted matter can be removed. If necessary, food areas shall be capable of being disinfected.

The food area comprises surfaces in contact with the food and surfaces that the food may contact during preparation of the product. The splash area comprises surfaces on which part of the food may splash or flow during normal use but this food does not become part of the product.

Compliance is checked by inspection after having operated the appliance as in normal use and then cleaning and disinfecting it in accordance with the instructions for maintenance.

22.113 Non-food areas that are not adequately separated from food areas of appliances that dispense food shall be constructed so that the retention of moisture or unwanted matter, and the ingress of vermin, is prevented. When this is unavoidable, the surfaces of the non-food areas shall be cleanable in accordance with **22.112**.

This requirement does not apply to splash areas and appliances that dispense food in sealed containers such as cans and bottles.

Compliance is checked by inspection.

22.114 An espresso coffee maker shall be constructed so that it is not possible to remove the coffee filter by a simple operation while there is a hazardous pressure within the container. This requirement is considered to be met if the coffee filter can only be removed after it has been rotated through an angle of at least 30°.

Compliance is checked by inspection and by manual test.

23 INTERNAL WIRING

This clause of Part 1 is applicable except as follows.

23.3 Modification:

The requirement also applies to maintenance operations.

The number of flexings is

- a) 200 000, for conductors flexed during normal use; and
- b) 10 000, for conductors flexed during maintenance operations.

23.101 Anchorages for internal wiring that can easily be replaced shall be constructed and located so that

- a) the wiring cannot touch the clamping screws of the anchorage if these screws are accessible, unless they are separated from accessible metal parts by supplementary insulation.
- b) the wiring is not clamped by a metal screw that bears directly on the wiring.
- c) for class I appliances, the anchorages are of insulating material or are provided with an insulating lining, unless failure of the insulation of the wiring does not make accessible metal parts live.
- d) for class II appliances, the anchorages are of insulating material, or if of metal, they are insulated from accessible metal parts by supplementary insulation.

Compliance is checked by inspection.

23.102 Internal wiring that is accessible in the maintenance area and is moved during normal operation shall comply with **25.13**, **25.14**, **25.15** and **25.21**.

Compliance is checked by the relevant tests.

24 COMPONENTS

This clause of Part 1 is applicable except as follows.

24.1.5 Addition:

For appliance couplers incorporating thermostats, thermal cut-outs or fuses in the connector, IS/IEC 60320-1 is applicable except that

- a) the earthing contact of the connector is allowed to be accessible, provided that this contact is not likely to be gripped during insertion or withdrawal of the connector.
- b) the temperature required for the test of **18** is that measured on the pins of the appliance inlet during the heating test of **11** of this standard.
- c) the breaking-capacity test of **19** is carried out using the inlet of the appliance.
- d) the temperature rise of current-carrying parts specified in **21** is not determined.

Thermal controls are not allowed in connectors complying with the standard sheets of IS/IEC 60320-1.

24.2 Modification:

Switches and automatic controls operating at safety extra-low voltage may be fitted in interconnection cords in the maintenance area.

24.101 Connecting devices of interconnection cords shall be identified if they are interchangeable with other connecting means in the appliance, if this could result in a hazard.

NOTE — Colour coding can be used for identification.

Compliance is checked by inspection.

24.102 Interlock switches shall comply with IS/IEC 61058-1 as far as is reasonable and shall ensure all-pole disconnection. However, single-pole disconnection is allowed for protection against mechanical hazards.

Compliance is checked by testing the switch in accordance with the relevant clauses of IS/IEC 61058-1, the number of cycles of operation for the test of **17** being 10 000. However, if the switch is operated once per delivery, the number of cycles of operation is 100 000. This requirement only applies to interlock switches necessary for compliance with this standard.

24.103 Thermal cut-outs incorporated for compliance with **19** shall not be self-resetting. They shall have a

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trip-free mechanism if they disconnect heating elements and if they disconnect motors, the unexpected starting of which may cause a hazard to the user or maintenance person.

Compliance is checked by inspection and by manual test.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CORDS

This clause of Part 1 is applicable as follows:

25.7 Addition:

Supply cords of appliances intended for outdoor use shall be polychloroprene sheathed and not be lighter than ordinary polychloroprene sheathed cord.

25.15 Addition:

When the test is carried out on internal wiring, the pull force is 30 N and the torque 0.1 Nm, irrespective of the mass of the appliance.

For internal wiring, a push force of 30 N is applied when pushing the wiring into the appliance.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of Part 1 is applicable.

27 PROVISION FOR EARTHING

This clause of Part 1 is applicable except as follows.

27.2 Addition:

Stationary class I appliances of the professional type intended to be installed in kitchens shall incorporate a terminal for the connection of an external equipotential bonding conductor. This terminal shall be connected to all accessible metal parts of the appliance and shall allow the connection of a conductor having a nominal cross-sectional area of 2.5 mm² to 10 mm². It shall be located so that the conductor can be connected after installation of the appliance. This requirement does not apply to small parts such as nameplates.

28 SCREWS AND CONNECTIONS

This clause of Part 1 is applicable except as follows.

28.1 Addition:

The requirement also applies to screws that may be removed during maintenance operations.

The test also applies to screws likely to be tightened during maintenance operations.

28.3 Addition:

The requirement also applies to screws operated by the maintenance person.

29 CLEARANCES, CREEPAGE DISTANCES AND SOLID INSULATION

This clause of Part 1 is applicable except as follows.

29.2 Addition:

The microenvironment is pollution degree 3 unless the insulation is enclosed or located so that it is unlikely to be exposed to pollution during normal use of the appliance due to

- a) condensation produced by the appliance, and
- b) the use of liquids and solids, such as ingredients, products or cleaning agents.

30 RESISTANCE TO HEAT AND FIRE

This clause of Part 1 is applicable except as follows:

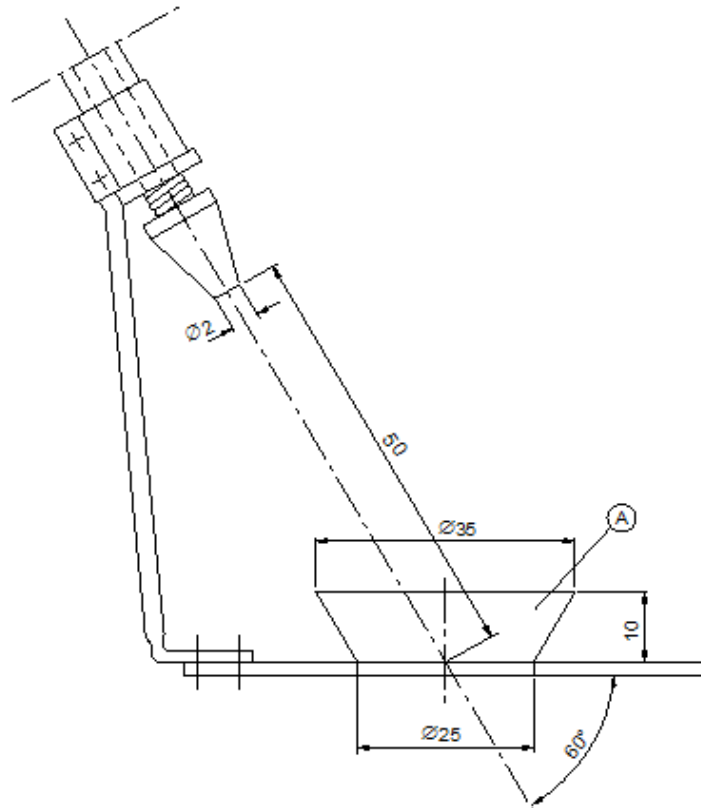
30.2.2 Not applicable.

31 RESISTANCE TO RUSTING

This clause of Part 1 is applicable.

32 RADIATION, TOXICITY AND SIMILAR HAZARDS

This clause of Part 1 is applicable.



All dimensions in millimetres.

Key

A — bowl

FIG. 101 SPLASH APPARATUS

101 TESTS

101.1 Type Tests

The tests specified in Table 101 shall constitute the

type tests and shall be carried out on a sample selected preferably at random from regular production lot (*see 5.3*). Before commencement of the tests, the water heater shall be visually examined and inspected of

Table 101 Schedule of Type Tests
(Clause 101.1)

SI No. (1)	Tests (2)	Ref to Clause (3)
i)	Protection against access to live parts	8
ii)	Power input and current	10
iii)	Heating	11
iv)	Leakage current and electric strength at operating temperature	13
v)	Transient over voltages	14
vi)	Moisture resistance	15
vii)	Leakage current and electric strength	16
viii)	Overload protection of transformers and associated circuits	17
ix)	Abnormal operation	19
x)	Stability and mechanical hazards	20
xi)	Mechanical strength	21
xii)	Construction	22
xiii)	Internal wiring	23
xiv)	Components	24
xv)	Supply connection and external flexible cords	25
xvi)	Terminals for external conductors	26
xvii)	Provision for earthing	27
xviii)	Screw and connections	28
xix)	Clearances, creepage distances and solid insulation	29
XX)	Resistance to heat and fire	30
xxi)	Resistance to rusting	31
xxii)	Radiation, toxicity and similar hazards	32

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components, parts and their assembly, constructions, mechanical hazards, marking provision of suitable terminals for supply connections, earthing and the effectiveness screws and connection. The external surface finish shall be even and free from finishing defects.

101.1.1 Criteria of Acceptance

Sample shall successfully pass all the type tests for proving conformity with the requirements of the standard. If the sample fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test (s) in which failure (s) had occurred. No failure should be permitted in the repeat test (s).

101.2 Acceptance Tests

The following shall constitute the acceptance tests:

<i>Sl No. (1)</i>	<i>Tests (2)</i>	<i>Ref to Clause (3)</i>
i)	Protection against access to live parts	08
ii)	Power input and current	10
iii)	Heating	11
iv)	Leakage current and electric strength at operating temperature	13
v)	Moisture resistance	15
vi)	Leakage current and electric strength	16
viii)	Provision for earthing	27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 h while conducting the test for moisture resistance (*see 15*).

ANNEXURE

The Annexures of Part 1 are applicable except as follows.

ANNEX AA

(Normative)

AGING TEST FOR ELASTOMERIC PARTS

The aging test on elastomeric parts is carried out by measuring their hardness and mass before and after immersion in water at elevated temperature.

The test is carried out on at least three samples of each part. The samples and test procedure are as specified in IS 3400 (Part 6) with the following modifications.

5 TEST LIQUIDS

The test is carried out with water.

Care is taken to ensure that the total mass of the test pieces immersed does not exceed 100 g for each litre of water, that the test pieces are completely immersed and that their entire surface is freely exposed to the water. During the tests, the test pieces are not exposed to direct light. Test pieces of different compounds are not immersed at the same time in the same solution.

6 TEST PIECES

6.4 Conditioning

The temperature is $23\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ and the relative

humidity is (50 ± 5) percent.

7 IMMERSION IN THE TEST LIQUID

7.1 Temperature

The water is heated within 1 h with the test pieces immersed, to a temperature of $75^{+5}_0\text{ }^{\circ}\text{C}$ and maintained at this value. Water at the same temperature is added to compensate for evaporation.

7.2 Duration

The test pieces are immersed for a total period of 48^{+1}_0 h .

The test pieces are then immediately immersed in fresh water that is maintained at ambient temperature. The pieces are immersed for $45\text{ min} \pm 15\text{ min}$.

After removal from the water, the test pieces are dried with blotting paper.

8 PROCEDURE

8.2 Change in Mass

The increase in mass of the test pieces shall not exceed 10 percent of the value determined before immersion.

8.6 Change in Hardness

The micro-test for hardness applies.

The hardness of the test pieces shall not have changed by more than 8 IRHD. Their surface shall not have become sticky and shall show no crack visible to the naked eye or any other deterioration.

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